

## STA4026Z 2026 – Statistics Honours Analytics

### Assignment 1 – Applied Supervised Learning

**Due Date: Monday 23 March 2026, 12:00 (noon)**

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**Please read the following instructions carefully:**

- You may complete this assignment either in pairs or individually.
  - Answer all questions using R. You may use any packages and functions you like.
  - You must submit the following on the course site:
    - A pdf report. **The page limit is 12 pages.** Anything beyond the 12<sup>th</sup> page will be ignored. The report should NOT contain any code.
    - All relevant R files (.R, .Rmd, .qmd, .RData). Be sure to use comments in your code and to set seeds to ensure reproducibility. Please **DO NOT ZIP** any files; upload them separately.
  - Use the following naming convention for the files:  
STDNUM001\_STA4026Z\_A1.pdf for individuals  
STDNUM001\_STDNUM002\_STA4026Z\_A1.pdf for pairs.
  - Your report may be compiled using any software you like, although you are strongly encouraged to either knit from RMarkdown or Quarto, or use L<sup>A</sup>T<sub>E</sub>X. Some marks will be awarded for communication and presentation, so make sure everything in your report is legible and neat.
  - You are welcome to include figures and tables not specifically asked for, but only do so if they are relevant to your discussion. Also be sure to interpret them sufficiently; marks will be deducted for any output placed in the report without discussion.
  - Do NOT paste/print raw R output verbatim – this will be penalised. If you want to include R output, typeset it properly or present it in a table.
  - To help the reader easily assimilate the information, round values to the fewest number of decimal places necessary (unless there is a good reason for expressing a more exact value).
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## Cape Town Airbnb listing analysis

It is well documented that short-term rental platforms such as Airbnb have had significant effects on housing affordability and long-term rental availability in Cape Town. While such platforms provide income opportunities for hosts and increase tourism flexibility, concerns persist regarding professional hosting, housing stock diversion, and price inflation in residential areas. In this assignment, you will model Airbnb listing data for Cape Town in order to predict one of a few possible binary outcomes related to potential housing-market impact.

You are provided with `listings.csv`, containing curated Airbnb listing data for Cape Town from 2025, obtained from [InsideAirbnb](#). This file must be read into R as is, i.e. the spreadsheets may not be edited at all. The feature descriptions are given in `variable_descriptions.csv`.

The data have been cleaned already and is ready for modelling. You do not *have* to use all the features, but if you omit any, this must be clearly communicated. If you want to (not necessary), you may perform additional preprocessing, as long as you **clearly document and report** what you did.

Provide a brief introduction to the report; a reader should be able to comprehend the task at hand without having seen this set of instructions. Although an exploration/description of the features is not required, a brief section on this may be included. However, if you do so, your discussion should focus only on aspects that pertain to this exercise.

The primary objective is not just to maximise predictive performance, but to demonstrate sound modelling procedure, careful hyperparameter tuning, appropriate validation strategy, and clear interpretation.

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First randomly split the data into training (80%) and testing (20%). The training set is to be used for all model fitting, tuning, and comparison. The test set is only for final model evaluation. Use 10-fold cross-validation when required to cross-validate.

### Question 1 – Creating a Target Variable [10 marks]

Your first task is to construct a binary target variable by choosing **one** (and only one) of the following definitions:

1.  $\text{HighPrice} = \begin{cases} 1 & \text{if } \text{price} \text{ exceeds the } x^{th} \text{ percentile.} \\ 0 & \text{otherwise} \end{cases}$
2.  $\text{HighHost} = \begin{cases} 1 & \text{if } \text{host\_listings\_count} > x. \\ 0 & \text{otherwise} \end{cases}$
3.  $\text{HighAvailability} = \begin{cases} 1 & \text{if } \text{availability\_365} > x \text{ days.} \\ 0 & \text{otherwise} \end{cases}$

Briefly state the reason for your choice. Whichever one you choose, decide on and motivate for a value of  $x$  using only the training set, and report the class imbalance of your target variable. Use this value of  $x$  to create the test set's target variable as well. You must of course remove the original version of the variable, but the other two original variables may still be used as features.

## Question 2 – Model Development and Selection [60 marks]

Using the training set, fit the following models. Remember to set seeds for reproducibility.

1. Regularised Elastic-net logistic regression. You may add non-linear terms if you wish.
2. A random forest.
3. Any boosted tree model (gbm, xgBoost, LightGBM,<sup>1</sup> etc.).
4. K-nearest neighbours.

For each model:

- a) Briefly summarise the modelling approach and hyperparameter selection process (and choice of features for KNN). Clearly state your hyperparameter search space before tuning. (40)
- b) Use cross-validation (CV) to evaluate each model using a decision threshold of  $\tau = 0.5$  and report the following metrics for each in a single table:  
Accuracy, F1 Score, Precision, Recall, Specificity, and AUROC (in that order). Also provide a visual illustration of the results. (15)
- c) Briefly compare the models and select (with justification) one as your final model. (5)

## Question 3 – Threshold Optimisation [30 marks]

Suppose the goal is to fit a model that will yield the highest F1 score on unseen data. Instead of finding new hyperparameters by using the F1 score as objective function, we will use your selected model and adjust the decision rule threshold ( $\tau$ ).

- a) For your selected model, plot the CV F1 score as a function of  $\tau$ , indicating the maximum value with a vertical line. (10)
- b) Plot the CV ROC curve. Indicate on the curve the point corresponding to the maximum F1 score, and add the F1 score and  $\tau$  as text next to the point. (10)
- c) Using this identified  $\tau$  value, calculate the same metrics from Question 2b) for the test set. Compare this with the CV results and briefly discuss. (10)

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<sup>1</sup>The documentation can be found [here](#).

## Question 4 – Interpretation and Error Analysis [30 marks]

For the selected model:

- a) Provide variable importance plots and interpret the most influential predictors. (10)
- b) Provide at most four partial dependence plots (your choice of how many) and interpret them in context. (10)

If the Elastic-net was your final model, replace a) and b) with a comprehensive interpretation of the retained variables' coefficients.

- c) Identify five false positives and five false negatives from your test set prediction. Are there any patterns you can identify in these misclassifications? (10)

$\mathcal{END}$

TOTAL MARKS = 130

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